Desflurane: Drug information - UpToDate

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Contributor Disclosures

For additional information see "Desflurane: Patient drug information" and "Desflurane: Pediatric drug information"

For abbreviations, symbols, and age group definitions show table

Brand Names: US

- Suprane

Pharmacologic Category

- General Anesthetic, Inhalation

Dosing: Adult

Note: Dosage must be individualized based on patient response.

Anesthesia, induction

Anesthesia, induction: Initial: Inhaled concentration of 3%, increased by 0.5% to 1% increments every 2 to 3 breaths (end tidal concentrations 4% to 11%). Inspired concentrations >12% have been safely administered during induction and may require a reduction of nitrous oxide or air.

Anesthesia, maintenance

Anesthesia, maintenance: Inhaled concentrations of 2.5% to 8.5% with or without concomitant nitrous oxide.

Mini mum alveolar concentration values:

The minimum alveolar concentration (MAC), the concentration at which 50% of patients do not respond to surgical incision, varies by age. In adults, the concentration at which amnesia and loss of awareness occur (MAC - awake) is 2.4%. Surgical levels of anesthesia are maintained between 2.5% to 8.5%.

Patient Age and Minimum Alveolar Concentration (MAC)

Age

MAC with 100% Oxygen

MAC with 60% N_2O/40% Oxygen

25 y

7.3%

4%

45 y

6%

2.8%

70 y

5.2%

1.7%

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult Drug Interactions database for more information.

Dosing: Kidney Impairment: Adult

No dosage adjustment necessary.

Dosing: Liver Impairment: Adult

No dosage adjustment necessary.

Dosing: Older Adult

The mean alveolar concentrations decrease with increasing age; adjust dose accordingly. Refer to adult dosing.

Dosing: Pediatric

(For additional information see "Desflurane: Pediatric drug information")

Anesthesia, maintenance

Anesthesia, maintenance: Note: The minimum alveolar concentration (MAC), the concentration at which 50% of patients do not respond to surgical incision, varies by age. Desflurane MAC decreases with increasing age; dosage must be individualized based on patient response; desflurane MAC may also be decreased with concomitant nitrous oxide administration.

Infants, Children, and Adolescents: Inhaled concentrations of ~7% to 10% and lower concentrations with nitrous oxide (4% to 7.5%).

Patient Age and Minimum Alveolar Concentration (MAC)

Age

MAC with 100% Oxygen

MAC with 60% N_2O/40% Oxygen

Note: Should only be administered with vaporizer specifically designed for desflurane.

10 weeks

9.4% ± 0.4%

9 months

10% ± 0.7%

7.5% ± 0.8%

2 years

9.1% ± 0.6%

3 years

6.4% ± 0.4%

4 years

8.6% ± 0.6%

7 years

8.1% ± 0.6%

25 years

7.3%

4% ± 0.3%

Dosage adjustment for concomitant therapy: Significant drug interactions exist, requiring dose/frequency adjustment or avoidance. Consult drug interactions database for more information.

Dosing: Kidney Impairment: Pediatric

No dosage adjustment necessary.

Dosing: Liver Impairment: Pediatric

No dosage adjustment necessary.

Adverse Reactions

The following adverse drug reactions and incidences are derived from product labeling unless otherwise specified. Adverse reactions reported for maintenance or recovery in pediatric patients and adults, unless otherwise indicated.

>10%:

Gastrointestinal: Nausea (27%), vomiting (16%)

Respiratory: Apnea (3% to 10%; induction [adults]: 15%), breath-holding (3% to 10%; induction [adults]: 30%), cough (induction [adults]: 34%)

1% to 10%:

Cardiovascular: Atrioventricular nodal arrhythmia (>1%), bradycardia (>1%), hypertension (>1%; including malignant hypertension), tachycardia (>1%)

Gastrointestinal: Sialorrhea (>1%)

Nervous system: Headache (>1%)

Neuromuscular & skeletal: Laryngospasm (3% to 10%)

Ophthalmic: Conjunctivitis (>1%)

Respiratory: Increased bronchial secretions (induction [adults]: 3% to 10%), increased cough (3% to 10%), oxygen saturation decreased (induction [adults]: 3% to 10%), pharyngitis (3% to 10%)

<1%:

Cardiovascular: Acute myocardial infarction, bigeminy, cardiac arrhythmia, ECG abnormality, ischemic heart disease, vasodilation

Dermatologic: Pruritus

Hematologic & oncologic: Hemorrhage

Nervous system: Agitation, dizziness

Neuromuscular & skeletal: Increased creatinine phosphokinase in blood specimen, myalgia

Respiratory: Asthma, dyspnea, hypoxia

Miscellaneous: Fever

Postmarketing:

Cardiovascular: Atrial fibrillation, hypotension, increased ST segment on ECG, inversion T wave on ECG, prolonged QT interval on ECG, shock, torsades de pointes, ventricular dysfunction

Dermatologic: Erythema of skin, urticaria

Endocrine & metabolic: Hyperammonemia, hyperkalemia, hypokalemia, metabolic acidosis

Gastrointestinal: Abdominal pain, acute pancreatitis, cholestasis

Hematologic & oncologic: Blood coagulation test abnormality, disorder of hemostatic components of blood

Hepatic: Hepatic failure, hepatic necrosis, hepatitis (including cytolytic hepatitis), hepatotoxicity (Chalasani 2021; Chung 2003; Côté 2007), increased serum bilirubin, increased serum transaminases (including increased serum alanine aminotransferase, increased serum aspartate aminotransferase), jaundice

Nervous system: Asthenia, delirium, malaise, seizure

Neuromuscular & skeletal: Hypokinesia, rhabdomyolysis

Ophthalmic: Scleral icterus

Respiratory: Bronchospasm, hemoptysis, respiratory distress, respiratory failure

Miscellaneous: Malignant hyperthermia

Contraindications

Hypersensitivity to desflurane, other halogenated agents, or any component of the formulation; known or suspected genetic susceptibility to malignant hyperthermia; patients in whom general anesthesia is contraindicated; induction of anesthesia in pediatric patients; history of moderate to severe hepatic impairment following anesthesia with desflurane or other halogenated agents and not otherwise explained.

Canadian labeling: Additional contraindications (not in US labeling): History of hepatitis due to a halogenated inhalational anesthetic or in whom hepatic dysfunction, jaundice or unexplained fever, leukocytosis, or eosinophilia has occurred after previous halogenated anesthetic administration.

Warnings/Precautions

Concerns related to adverse effects:

• Decreased blood flow: May cause decrease in hepatic and/or renal blood flow.

• Hepatitis: May cause sensitivity hepatitis in patients who have been sensitized by previous exposure to halogenated anesthetics.

• Hyperkalemia: Use of inhaled anesthetics has been associated with rare cases of perioperative hyperkalemia (including fatalities) in pediatric patients; concomitant use of succinylcholine was associated with many of the reported cases, but not all. Risk of hyperkalemia is increased in patients with underlying neuromuscular disease (eg, Duchenne muscular dystrophy). Other abnormalities may include elevation in CK and myoglobinuria. Monitor closely for arrhythmias. Aggressively identify and treat hyperkalemia.

• Increased intracranial pressure: May dilate the cerebral vasculature and may, in certain conditions, increase intracranial pressure. In patients with intracranial space-occupying lesions, administer at ≤0.8 MAC in conjunction with a barbiturate induction and hyperventilation in the period before cranial decompression; maintain cerebral perfusion pressure.

• Malignant hyperthermia: May trigger malignant hyperthermia; some reported cases have been fatal. Risk may be increased with concomitant administration of succinylcholine and volatile anesthetic agents and patients with genetic factors or family history of malignant hyperthermia, including ryanodine receptor or dihydropyridine receptor inherited variants. Signs of malignant hyperthermia may include arrhythmias, cyanosis, hemodynamic instability, hypercapnia, hyperthermia, hypovolemia, hypoxia, muscle rigidity, tachycardia, and tachypnea; coagulopathies, renal failure, and skin mottling may also occur. If malignant hyperthermia is suspected, discontinue triggering agents and institute appropriate therapy (eg, dantrolene) and other supportive measures.

• QT prolongation: Cases of QT prolongation in association with torsade de pointes (some fatal) have been reported with inhaled anesthetic agents; use caution when administering to patients at risk of QT prolongation (eg, concurrent use of drugs that can prolong the QT interval, such as class Ia and III antiarrhythmic drugs, elderly patients, congenital QT prolongation) (Han 2010; Kang 2006; Nakao 2010).

• Respiratory depression: Causes dose-dependent respiratory depression and blunted ventilatory response to hypoxia and hypercapnia. Hypoxic pulmonary vasoconstriction is blunted which may lead to increased pulmonary shunt. May produce elevated carbon monoxide levels and carboxyhemoglobin in the presence of a desiccated dry carbon dioxide absorbent within the circle breathing system of an anesthetic machine; barium hydroxide and soda lime desiccation have been reported when fresh gasses are passed over a carbon dioxide canister at high flow rates over many hours or days. Maintain fresh absorbent as per manufacturer guidelines regardless of state of colorimetric indicator.

Disease-related concerns:

• Cardiovascular disease: Do not use as a single agent to induce anesthesia in patients with CAD or in whom an increase in heart rate or blood pressure should be avoided. Abrupt increases in inspired concentrations >1 MAC can produce a transient increase in blood pressure and heart rate due to increased plasma catecholamine levels. Acute, rapid increases in desflurane concentration can produce increased sympathetic cardiovascular stimulation which lasts for 2 to 4 minutes; can be blunted by concurrent use of nitrous oxide, opioids, beta-blockers, and alpha-2 agonists (Weiskopf 1994). Hypotensive effect due to peripheral vasodilation is dose dependent and increases as anesthesia is deepened. In a scientific statement from the American Heart Association, desflurane has been determined to be an agent that may exacerbate underlying myocardial dysfunction (magnitude: major) (AHA [Page 2016]).

• Hepatic disease: Due to the risk of hepatitis with halogenated anesthetics, use in patients with cirrhosis, viral hepatitis, or other preexisting hepatic disease should be approached with caution; disruption of hepatic function, icterus, and fatal liver necrosis have been reported and may indicate hypersensitivity. Consider the use of an anesthetic other than a halogenated anesthetic.

Special populations:

• Pediatric: Postoperative agitation may occur in children while emerging from anesthesia. Contraindicated for induction of general anesthesia in pediatric patients due to the higher incidence of moderate to severe upper airway adverse events (eg, laryngospasm, coughing, breath-holding, increased secretions). Do not use to maintain anesthesia in nonintubated pediatric patients due to a high incidence of moderate to severe upper airway adverse events. Use with caution when used to maintain anesthesia in children ≤6 years when a laryngeal mask airway (LMA) is in place due to the increased potential for adverse respiratory effects especially with removal of the LMA under deep anesthesia. Use with caution in children with asthma or recent upper respiratory infection; increased risk for airway narrowing and increased airway resistance.

• Pediatric neurotoxicity: In pediatric and neonatal patients <3 years and patients in third trimester of pregnancy (ie, times of rapid brain growth and synaptogenesis), the repeated or lengthy exposure to sedatives or anesthetics during surgery/procedures may have detrimental effects on child or fetal brain development and may contribute to various cognitive and behavioral problems. Epidemiological studies in humans have reported various cognitive and behavioral problems including neurodevelopmental delay (and related diagnoses), learning disabilities, and ADHD. Human clinical data suggest that single, relatively short exposures are not likely to have similar negative effects. No specific anesthetic/sedative has been found to be safer. For elective procedures, risk vs benefits should be evaluated and discussed with parents/caregivers/patients; critical surgeries should not be delayed (FDA 2016).

Special handling:

• Occupational caution: Although there are no documented adverse effects of chronic occupational exposure to halogenated anesthetic vapors, like desflurane, some epidemiological studies suggest a link between these anesthetics and increased health problems (particularly spontaneous abortion). There is no specific work exposure limit established for desflurane. However, the National Institute for Occupational Safety and Health (NIOSH) has recommended that workers should not be exposed to ceiling concentrations greater than 2 ppm of any halogenated anesthetic agent over a period ≤1 hour. The predicted effects of acute overexposure by inhalation of desflurane include headache, dizziness, or (in extreme cases) unconsciousness. Precautions (eg, adequate ventilation, scavenging-systems, minimizing leaks/spills) can help to lessen any potential risk.

Other warnings/precautions:

• Monitoring: Transient elevations in white blood cell count and glucose may occur.

• Vaporizer use: Yellow discoloration of desflurane has been observed through the vaporizer sight glass or after draining the vaporizer; quality or efficacy of desflurane is not altered in these situations. Refer to the vaporizer Instructions For Use or contact manufacturer for recommended actions.

Dosage Forms: US

Excipient information presented when available (limited, particularly for generics); consult specific product labeling.

Solution, Inhalation:

Suprane: (240 mL); 100% (240 mL)

Generic: (240 mL)

Generic Equivalent Available: US

Yes

Administration: Adult

Inhalation: Via desflurane-specific calibrated heated vaporizer

Administration: Pediatric

Inhalation: Administer via desflurane-specific calibrated heated vaporizer.

Storage/Stability

Store at 15°C to 30°C (59°F to 86°F).

Use: Labeled Indications

Anesthesia: For induction of anesthesia for inpatient and outpatient surgery and maintenance of anesthesia in adults. For maintenance of anesthesia for inpatient and outpatient surgery in pediatric patients following induction with other agents and tracheal intubation. **Note:** Use of desflurane for induction of general anesthesia in adult patients may not be suitable due to its irritant properties and unpleasant odor which may cause coughing, breath-holding, laryngospasm, oxygen desaturation, increased secretions, hypertension, and tachycardia.

Medication Safety Issues

Sound-alike/look-alike issues:

Desflurane may be confused with Desferal

High alert medication:

The Institute for Safe Medication Practices (ISMP) includes this medication among its list of drug classes (anesthetic agent, general, inhaled and IV) which have a heightened risk of causing significant patient harm when used in error (ISMP List of High-Alert Medications in Acute Care Settings ©ISMP 2024).

Metabolism/Transport Effects

None known.

Drug Interactions

(For additional information: Launch drug interactions program)

Note: Interacting drugs may **not be individually listed below** if they are part of a group interaction (eg, individual drugs within "CYP3A4 Inducers [Strong]" are NOT listed). For a complete list of drug interactions by individual drug name and detailed management recommendations, use the drug interactions program by clicking on the "Launch drug interactions program" link above.

Acrivastine: May increase adverse/toxic effects of Inhalational Anesthetics. *Risk C: Monitor*

Acrivastine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Alcohol (Ethyl): CNS Depressants may increase CNS depressant effects of Alcohol (Ethyl). *Risk C: Monitor*

Alfuzosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Alizapride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Allopurinol: CNS Depressants may increase sedative effects of Allopurinol. *Risk C: Monitor*

Amifostine: Blood Pressure Lowering Agents may increase hypotensive effects of Amifostine. Management: When used at chemotherapy doses, hold blood pressure lowering medications for 24 hours before amifostine administration. If blood pressure lowering therapy cannot be held, do not administer amifostine. Use caution with radiotherapy doses of amifostine. *Risk D: Consider Therapy Modification*

Amisulpride (Oral): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amisulpride (Oral): May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Amisulpride (Oral): May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk C: Monitor*

Amitriptyline: CNS Depressants may increase CNS depressant effects of Amitriptyline. *Risk C: Monitor*

Amitriptylinoxide: CNS Depressants may increase CNS depressant effects of Amitriptylinoxide. *Risk C: Monitor*

Amoxapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Amphetamines: May increase hypotensive effects of Inhalational Anesthetics. *Risk C: Monitor*

Antipsychotic Agents (Second Generation [Atypical]): Blood Pressure Lowering Agents may increase hypotensive effects of Antipsychotic Agents (Second Generation [Atypical]). *Risk C: Monitor*

Arginine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Asenapine: CNS Depressants may increase CNS depressant effects of Asenapine. *Risk C: Monitor*

Azelastine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Baclofen: CNS Depressants may increase CNS depressant effects of Baclofen. *Risk C: Monitor*

Bambuterol: May increase arrhythmogenic effects of Inhalational Anesthetics. Management: Some labels recommend specifically avoiding halothane; others recommend separating administration by at least 6 hours; other bambuterol labels do not mention this possible interaction. Monitor for increased sensitivity to arrhythmias if coadministered. *Risk D: Consider Therapy Modification*

Barbiturates: CNS Depressants may increase CNS depressant effects of Barbiturates. *Risk C: Monitor*

Barbiturates: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Benperidol: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Benzodiazepines: CNS Depressants may increase CNS depressant effects of Benzodiazepines. *Risk C: Monitor*

Blonanserin: CNS Depressants may increase CNS depressant effects of Blonanserin. Management: Use caution if coadministering blonanserin and CNS depressants; dose reduction of the other CNS depressant may be required. Strong CNS depressants should not be coadministered with blonanserin. *Risk D: Consider Therapy Modification*

Blood Pressure Lowering Agents: May increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Brexanolone: CNS Depressants may increase CNS depressant effects of Brexanolone. *Risk C: Monitor*

Brimonidine (Ophthalmic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Brimonidine (Topical): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Brivaracetam: CNS Depressants may increase CNS depressant effects of Brivaracetam. *Risk C: Monitor*

Bromopride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Bromperidol: May decrease hypotensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase hypotensive effects of Bromperidol. *Risk X: Avoid*

Brompheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buclizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Buprenorphine: CNS Depressants may increase CNS depressant effects of Buprenorphine. Management: Avoid coadministration of buprenorphine and other CNS depressants if possible. If combined, consider adjustments to induction, increased monitoring, and co-prescription of an opioid overdose reversal agent. *Risk D: Consider Therapy Modification*

BusPIRone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Calcium Channel Blockers: Inhalational Anesthetics may increase hypotensive effects of Calcium Channel Blockers. *Risk C: Monitor*

Cannabinoid-Containing Products: CNS Depressants may increase CNS depressant effects of Cannabinoid-Containing Products. *Risk C: Monitor*

CarBAMazepine: CNS depressant effects of CarBAMazepine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Carbinoxamine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Carisoprodol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Cenobamate: CNS depressant effects of Cenobamate and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cetirizine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk D: Consider Therapy Modification*

Chloral Hydrate-Chloral Betaine: CNS Depressants may increase CNS depressant effects of Chloral Hydrate-Chloral Betaine. Management: Consider alternatives to the use of chloral hydrate or chloral betaine and additional CNS depressants. If combined, consider a dose reduction of either agent and monitor closely for enhanced CNS depressive effects. *Risk D: Consider Therapy Modification*

Chlormethiazole: May increase CNS depressant effects of CNS Depressants. Management: Monitor closely for evidence of excessive CNS depression. The chlormethiazole labeling states that an appropriately reduced dose should be used if such a combination must be used. *Risk D: Consider Therapy Modification*

Chlorphenesin Carbamate: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Chlorpheniramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

ChlorproMAZINE: CNS depressant effects of ChlorproMAZINE and CNS Depressants may increase with coadministration. Management: If possible, discontinue the use of other CNS depressants prior to initiating chlorpromazine; other CNS depressants can be re-started at low doses and titrated slowly as needed. Monitor for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Chlorprothixene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Chlorzoxazone: CNS depressant effects of Chlorzoxazone and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cinnarizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clemastine: CNS depressant effects of Clemastine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

ClomiPRAMINE: CNS depressant effects of ClomiPRAMINE and CNS Depressants may increase with coadministration. *Risk C: Monitor*

CloNIDine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Clothiapine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

CloZAPine: CNS Depressants may increase CNS depressant effects of CloZAPine. *Risk C: Monitor*

CNS Depressants: May increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Cyclizine: CNS depressant effects of Cyclizine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Cyclobenzaprine: CNS depressant effects of CNS Depressants and Cyclobenzaprine may increase with coadministration. *Risk C: Monitor*

Cyproheptadine: CNS depressant effects of Cyproheptadine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Dabrafenib: QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of Dabrafenib. Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Dantrolene: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Daridorexant: May increase CNS depressant effects of CNS Depressants. Management: Dose reduction of daridorexant and/or any other CNS depressant may be necessary. Use of daridorexant with alcohol is not recommended, and the use of daridorexant with any other drug to treat insomnia is not recommended. *Risk D: Consider Therapy Modification*

Desipramine: CNS Depressants may increase CNS depressant effects of Desipramine. *Risk C: Monitor*

Deutetrabenazine: CNS depressant effects of CNS Depressants and Deutetrabenazine may increase with coadministration. *Risk C: Monitor*

Dexbrompheniramine: CNS Depressants may increase CNS depressant effects of Dexbrompheniramine. *Risk C: Monitor*

Dexchlorpheniramine: CNS depressant effects of CNS Depressants and Dexchlorpheniramine may increase with coadministration. *Risk C: Monitor*

DexmedeTOMIDine: CNS Depressants may increase CNS depressant effects of DexmedeTOMIDine. Management: Monitor for increased CNS depression during coadministration of dexmedetomidine and CNS depressants, and consider dose reductions of either agent to avoid excessive CNS depression. *Risk D: Consider Therapy Modification*

Dexmethylphenidate-Methylphenidate: May increase hypertensive effects of Inhalational Anesthetics. Management: Avoid the use of dexmethylphenidate/methylphenidate on the day of a planned surgery with an inhalational anesthetic. *Risk D: Consider Therapy Modification*

Diazoxide (Immediate Release): May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Difelikefalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Difenoxin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Dihydralazine: CNS Depressants may increase hypotensive effects of Dihydralazine. *Risk C: Monitor*

Dimenhydrinate: CNS depressant effects of CNS Depressants and Dimenhydrinate may increase with coadministration. *Risk C: Monitor*

Dimethindene (Systemic): CNS Depressants may increase CNS depressant effects of Dimethindene (Systemic). *Risk C: Monitor*

Dimethindene (Topical): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

DiphenhydrAMINE (Systemic): CNS depressant effects of CNS Depressants and DiphenhydrAMINE (Systemic) may increase with coadministration. *Risk C: Monitor*

Diphenoxylate: May increase CNS depressant effects of CNS Depressants. Management: Avoid coadministration of diphenoxylate/atropine with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients closely for CNS depression (eg, sedation, somnolence) with any combined use. *Risk D: Consider Therapy Modification*

Domperidone: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Domperidone. *Risk X: Avoid*

DOPamine: Inhalational Anesthetics may increase arrhythmogenic effects of DOPamine. *Risk C: Monitor*

Dothiepin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Doxapram: May increase adverse/toxic effects of Inhalational Anesthetics. Management: Delay administration of doxapram until the inhalational anesthetic has been excreted (ie, after 5 half-lives) in order to lessen the potential for arrhythmias, including ventricular tachycardia and ventricular fibrillation. *Risk D: Consider Therapy Modification*

Doxepin (Systemic): CNS depressant effects of CNS Depressants and Doxepin (Systemic) may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Doxepin (Systemic). Specifically, the risk of abnormal thinking or performing complex behaviors while asleep may be increased. Management: Monitor patients for CNS depression (eg, sedation, somnolence) if doxepin is coadministered with CNS depressants. If patients experience abnormal thinking or perform complex behaviors while asleep (eg, driving) strongly consider discontinuing doxepin. *Risk C: Monitor*

Doxepin (Topical): CNS Depressants may increase CNS depressant effects of Doxepin (Topical). *Risk C: Monitor*

Doxylamine: CNS Depressants may increase CNS depressant effects of Doxylamine. *Risk C: Monitor*

DroPERidol: May increase CNS depressant effects of CNS Depressants. Management: Consider dose reductions of droperidol or of other CNS agents (eg, opioids, barbiturates) with concomitant use. *Risk D: Consider Therapy Modification*

DroPERidol: May increase CNS depressant effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). DroPERidol may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics

(Moderate Risk). Management: Consider dose reductions and monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

DULoxetine: Blood Pressure Lowering Agents may increase hypotensive effects of DULoxetine. *Risk C: Monitor*

Emedastine (Systemic): May increase CNS depressant effects of CNS Depressants. Management: Consider avoiding this combination if possible. If required, monitor for excessive sedation or CNS depression, limit the dose and duration of combination therapy, and consider CNS depressant dose reductions. *Risk C: Monitor*

Entacapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ephedra: May increase arrhythmogenic effects of Inhalational Anesthetics. *Risk C: Monitor*

EPHEDrine (Nasal): May increase arrhythmogenic effects of Inhalational Anesthetics. *Risk X: Avoid*

EPINEPHrine (Systemic): Inhalational Anesthetics may increase arrhythmogenic effects of EPINEPHrine (Systemic). Management: Administer epinephrine with added caution in patients receiving, or who have recently received, inhalational anesthetics. Use lower than normal doses of epinephrine and monitor for the development of cardiac arrhythmias. *Risk D: Consider Therapy Modification*

Esketamine (Injection): CNS Depressants may increase therapeutic effects of Esketamine (Injection). Specifically, duration of esketamine injection may be prolonged. CNS Depressants may decrease adverse/toxic effects of Esketamine (Injection). *Risk C: Monitor*

Esketamine (Injection): May increase sedative effects of Inhalational Anesthetics. *Risk C: Monitor*

Esketamine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Eszopiclone: CNS depressant effects of CNS Depressants and Eszopiclone may increase with coadministration. CNS Depressants may increase adverse/toxic effects of Eszopiclone. Specifically, the risk of daytime psychomotor impairment may be increased. Management: Consider reducing the dose of eszopiclone or other CNS depressants if these agents are used concomitantly, and monitor for excessive daytime psychomotor impairment with any combined use. Use of other sedative-hypnotics during the night is not recommended. *Risk D: Consider Therapy Modification*

Fenfluramine: CNS Depressants may increase CNS depressant effects of Fenfluramine. *Risk C: Monitor*

Fenfluramine: May increase hypotensive effects of Inhalational Anesthetics. *Risk C: Monitor*

Fenoterol: Inhalational Anesthetics may increase arrhythmogenic effects of Fenoterol. *Risk C: Monitor*

Flibanserin: CNS Depressants may increase CNS depressant effects of Flibanserin. *Risk C: Monitor*

Flunarizine: CNS Depressants may increase CNS depressant effects of Flunarizine. *Risk X: Avoid*

Fluorouracil Products: QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of Fluorouracil Products. Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Flupentixol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

FluPHENAZine: CNS Depressants may increase CNS depressant effects of FluPHENAZine. *Risk C: Monitor*

Gabapentin (Immediate Release): CNS Depressants may increase CNS depressant effects of Gabapentin (Immediate Release). *Risk C: Monitor*

Gabapentin Enacarbil (Extended Release): CNS Depressants may increase CNS depressant effects of Gabapentin Enacarbil (Extended Release). *Risk C: Monitor*

Ganaxolone: CNS Depressants may increase CNS depressant effects of Ganaxolone. *Risk C: Monitor*

GuanFACINE: CNS Depressants may increase CNS depressant effects of GuanFACINE. *Risk C: Monitor*

Haloperidol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Haloperidol: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Herbal Products with Blood Pressure Lowering Effects: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

HydrOXYzine: May increase CNS depressant effects of CNS Depressants. Management: Reduce the dose of hydroxyzine or other CNS depressants if these agents are combined due to the potential for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Hypotension-Associated Agents: Blood Pressure Lowering Agents may increase hypotensive effects of Hypotension-Associated Agents. *Risk C: Monitor*

Iloperidone: CNS Depressants may increase CNS depressant effects of Iloperidone. *Risk C: Monitor*

Iloperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Imipramine: CNS Depressants may increase CNS depressant effects of Imipramine. *Risk C: Monitor*

Inhalational Anesthetics: CNS Depressants may increase CNS depressant effects of Inhalational Anesthetics. *Risk C: Monitor*

Isocarboxazid: CNS Depressants may increase adverse/toxic effects of Isocarboxazid. *Risk X: Avoid*

Isocarboxazid: May increase hypotensive effects of Inhalational Anesthetics. *Risk X: Avoid*

Isoproterenol: Inhalational Anesthetics may increase arrhythmogenic effects of Isoproterenol. *Risk X: Avoid*

Ixabepilone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kava Kava: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Ketamine: CNS Depressants may increase CNS depressant effects of Ketamine. *Risk C: Monitor*

Ketotifen (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Kratom: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Lacosamide: CNS Depressants may increase CNS depressant effects of Lacosamide. *Risk C: Monitor*

Lasmiditan: CNS Depressants may increase CNS depressant effects of Lasmiditan. *Risk C: Monitor*

Lemborexant: May increase CNS depressant effects of CNS Depressants. Management: Dosage adjustments of lemborexant and of concomitant CNS depressants may be necessary when administered together because of potentially additive CNS depressant effects. Close monitoring for CNS depressant effects is necessary. *Risk D: Consider Therapy Modification*

LevETIRAcetam: CNS Depressants may increase CNS depressant effects of LevETIRAcetam. *Risk C: Monitor*

Levocetirizine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Levodopa-Foslevodopa: Blood Pressure Lowering Agents may increase hypotensive effects of Levodopa-Foslevodopa. *Risk C: Monitor*

Levoketoconazole: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Levoketoconazole. *Risk X: Avoid*

Lisuride: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofepramine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lofexidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Lormetazepam: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Loxapine: CNS Depressants may increase CNS depressant effects of Loxapine. Management: Consider reducing the dose of CNS depressants administered concomitantly with loxapine due to an increased risk of respiratory depression, sedation, hypotension, and syncope. *Risk D: Consider Therapy Modification*

Lumateperone: CNS Depressants may increase CNS depressant effects of Lumateperone. *Risk C: Monitor*

Lurasidone: CNS Depressants may increase CNS depressant effects of Lurasidone. *Risk C: Monitor*

Magnesium Sulfate: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meclizine: CNS depressant effects of CNS Depressants and Meclizine may increase with coadministration. *Risk C: Monitor*

Melitracen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Meprobamate: CNS depressant effects of CNS Depressants and Meprobamate may increase with coadministration. *Risk C: Monitor*

Mequitazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Metaraminol: Inhalational Anesthetics may increase arrhythmogenic effects of Metaraminol. *Risk X: Avoid*

Metaxalone: CNS depressant effects of CNS Depressants and Metaxalone may increase with coadministration. *Risk C: Monitor*

Metergoline: May decrease antihypertensive effects of Blood Pressure Lowering Agents. Blood Pressure Lowering Agents may increase orthostatic hypotensive effects of Metergoline. *Risk C: Monitor*

Metergoline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methadone: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase CNS depressant effects of Methadone. Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation, ventricular arrhythmias, sedation, and respiratory depression. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

Methocarbamol: CNS Depressants may increase CNS depressant effects of Methocarbamol. *Risk C: Monitor*

Methotrimeprazine: CNS Depressants may increase CNS depressant effects of Methotrimeprazine. Methotrimeprazine may increase CNS depressant effects of CNS Depressants. Management: Reduce the usual dose of CNS depressants by 50% if starting methotrimeprazine until the dose of methotrimeprazine is stable. Monitor patient closely for evidence of CNS depression. *Risk D: Consider Therapy Modification*

Methoxyflurane: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Methsuximide: CNS Depressants may increase CNS depressant effects of Methsuximide. *Risk C: Monitor*

Metoclopramide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

MetyroSINE: CNS Depressants may increase sedative effects of MetyroSINE. *Risk C: Monitor*

Mianserin: CNS Depressants may increase CNS depressant effects of Mianserin. *Risk C: Monitor*

Milsaperidone: CNS Depressants may increase CNS depressant effects of Milsaperidone. *Risk C: Monitor*

Milsaperidone: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Minocycline (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Mirogabalin: CNS Depressants may increase CNS depressant effects of Mirogabalin. *Risk C: Monitor*

Mirtazapine: CNS Depressants may increase CNS depressant effects of Mirtazapine. *Risk C: Monitor*

Molindone: CNS Depressants may increase CNS depressant effects of Molindone. *Risk C: Monitor*

Molsidomine: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Moxonidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nabilone: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Naftopidil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nalfurafine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Nefopam: CNS Depressants may increase CNS depressant effects of Nefopam. *Risk C: Monitor*

Neuromuscular-Blocking Agents (Nondepolarizing): Inhalational Anesthetics may increase neuromuscular-blocking effects of Neuromuscular-Blocking Agents (Nondepolarizing). Management: When initiating a non-depolarizing neuromuscular blocking agent (NMBA) in a patient receiving an inhalational anesthetic, initial NMBA doses should be reduced 15% to 25% and doses of continuous infusions should be reduced 30% to 60%. *Risk D: Consider Therapy Modification*

Nicergoline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nicorandil: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Nitroprusside: Blood Pressure Lowering Agents may increase hypotensive effects of Nitroprusside. *Risk C: Monitor*

Nitrous Oxide: CNS Depressants may increase CNS depressant effects of Nitrous Oxide. *Risk C: Monitor*

Norepinephrine: Inhalational Anesthetics may increase arrhythmogenic effects of Norepinephrine. *Risk C: Monitor*

Nortriptyline: CNS Depressants may increase CNS depressant effects of Nortriptyline. *Risk C: Monitor*

Noscapine: CNS Depressants may increase adverse/toxic effects of Noscapine. *Risk X: Avoid*

Obinutuzumab: May increase hypotensive effects of Blood Pressure Lowering Agents. Management: Consider temporarily withholding blood pressure lowering medications beginning 12 hours prior to obinutuzumab infusion and continuing until 1 hour after the end of the infusion. *Risk D: Consider Therapy Modification*

OLANZapine: CNS depressant effects of OLANZapine and CNS Depressants may increase with coadministration. *Risk C: Monitor*

Olopatadine (Nasal): May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Ondansetron: May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Opicapone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Opioid Agonists: CNS Depressants may increase CNS depressant effects of Opioid Agonists. Management: Avoid concomitant use of opioid agonists and benzodiazepines or other CNS depressants when possible. These agents should only be combined if alternative treatment options are inadequate. If combined, limit the dosage and duration of each drug. *Risk D: Consider Therapy Modification*

Opipramol: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Orphenadrine: CNS Depressants may increase CNS depressant effects of Orphenadrine. *Risk C: Monitor*

Oxatomide: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Oxomemazine: May increase CNS depressant effects of CNS Depressants. *Risk X: Avoid*

Oxybate (Mixed Salts and Sodium Salt): CNS Depressants may increase CNS depressant effects of Oxybate (Mixed Salts and Sodium Salt). Management: Consider alternatives to this combination when possible. If combined, dose reduction or discontinuation of one or more CNS depressants (including the oxybate salt product) should be considered. Interrupt oxybate salt treatment during short-term opioid use *Risk D: Consider Therapy Modification*

Paliperidone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Paraldehyde: CNS Depressants may increase CNS depressant effects of Paraldehyde. *Risk X: Avoid*

Pentamidine (Systemic): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Pentoxifylline: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Perampanel: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Periciazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Perphenazine: CNS depressant effects of CNS Depressants and Perphenazine may increase with coadministration. *Risk C: Monitor*

Pheniramine: CNS Depressants may increase CNS depressant effects of Pheniramine. *Risk C: Monitor*

Phenylephrine (Ophthalmic): May increase adverse/toxic effects of Inhalational Anesthetics. Specifically, the cardiovascular depressant effects of inhalational anesthetics may be increased. *Risk C: Monitor*

Phenyltoloxamine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pholcodine: Blood Pressure Lowering Agents may increase hypotensive effects of Pholcodine. *Risk C: Monitor*

Pholcodine: CNS Depressants may increase CNS depressant effects of Pholcodine. *Risk C: Monitor*

Phosphodiesterase 5 Inhibitors: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Pimozide: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Pipamperone: May increase adverse/toxic effects of CNS Depressants. *Risk C: Monitor*

Piperaquine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Piperaquine. *Risk X: Avoid*

Piribedil: CNS Depressants may increase CNS depressant effects of Piribedil. *Risk C: Monitor*

Pizotifen: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Pomalidomide: CNS Depressants may increase CNS depressant effects of Pomalidomide. *Risk C: Monitor*

Pramipexole: CNS Depressants may increase sedative effects of Pramipexole. *Risk C: Monitor*

Pregabalin: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Procarbazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Prochlorperazine: CNS Depressants may increase CNS depressant effects of Prochlorperazine. *Risk C: Monitor*

Promazine: CNS Depressants may increase CNS depressant effects of Promazine. *Risk C: Monitor*

Promethazine: CNS Depressants may increase CNS depressant effects of Promethazine. Management: Discontinue or reduce the dose of other CNS depressants during promethazine therapy. *Risk D: Consider Therapy Modification*

Propofol: CNS Depressants may increase CNS depressant effects of Propofol. *Risk C: Monitor*

Prostacyclin Analogues: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Protriptyline: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

PyRIDostigmine: May increase or decrease neuromuscular-blocking effects of Inhalational Anesthetics. *Risk C: Monitor*

Pyrilamine (Systemic): May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

QT-prolonging Agents (Highest Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Agents (Highest Risk). Management: Consider alternatives to this drug combination. If combined, monitor for QTc interval prolongation and ventricular arrhythmias. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk D: Consider Therapy Modification*

QT-prolonging Antidepressants (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Antidepressants (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Class IC Antiarrhythmics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Class IC Antiarrhythmics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-Prolonging Inhalational Anesthetics (Moderate Risk): May increase hypotensive effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for hypotension and QTc interval prolongation and ventricular arrhythmias, including torsades de pointes when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Kinase Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Miscellaneous Agents (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Moderate CYP3A4 Inhibitors (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Quinolone Antibiotics (Moderate Risk): QT-Prolonging Inhalational Anesthetics (Moderate Risk) may increase QTc-prolonging effects of QT-prolonging Quinolone Antibiotics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

QT-prolonging Strong CYP3A4 Inhibitors (Moderate Risk): May increase QTc-prolonging effects of QT-Prolonging Inhalational Anesthetics (Moderate Risk). Management: Monitor for QTc interval prolongation and ventricular arrhythmias when these agents are combined. Patients with additional risk factors for QTc prolongation may be at even higher risk. *Risk C: Monitor*

Ramelteon: CNS Depressants may increase CNS depressant effects of Ramelteon. *Risk C: Monitor*

Rasagiline: Blood Pressure Lowering Agents may increase hypotensive effects of Rasagiline. *Risk C: Monitor*

Reproterol: Inhalational Anesthetics may increase arrhythmogenic effects of Reproterol. *Risk C: Monitor*

Reserpine: CNS Depressants may increase CNS depressant effects of Reserpine. *Risk C: Monitor*

Rilmenidine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

RisperiDONE: CNS depressant effects of CNS Depressants and RisperiDONE may increase with coadministration. *Risk C: Monitor*

Ritodrine: May increase adverse/toxic effects of Inhalational Anesthetics. *Risk C: Monitor*

Ropeginterferon Alfa-2b: CNS Depressants may increase adverse/toxic effects of Ropeginterferon Alfa-2b. Specifically, the risk of neuropsychiatric adverse effects may be increased. Management: Avoid coadministration of ropeginterferon alfa-2b and other CNS depressants. If this combination cannot be avoided, monitor patients for neuropsychiatric adverse effects (eg, depression, suicidal ideation, aggression, mania). *Risk D: Consider Therapy Modification*

ROPINIRole: CNS Depressants may increase sedative effects of ROPINIRole. *Risk C: Monitor*

Rotigotine: CNS Depressants may increase sedative effects of Rotigotine. *Risk C: Monitor*

Scopolamine: CNS Depressants may increase CNS depressant effects of Scopolamine. Management: Avoid coadministration of scopolamine with other CNS depressants if possible. If coadministration is required, monitor patients for excessive CNS depression. *Risk D: Consider Therapy Modification*

Selegiline: Blood Pressure Lowering Agents may increase hypotensive effects of Selegiline. *Risk C: Monitor*

Sertindole: May increase QTc-prolonging effects of QT-prolonging Agents (Moderate Risk). *Risk X: Avoid*

Silodosin: May increase hypotensive effects of Blood Pressure Lowering Agents. *Risk C: Monitor*

Stiripentol: CNS Depressants may increase CNS depressant effects of Stiripentol. *Risk C: Monitor*

Succinylcholine: Desflurane may increase neuromuscular-blocking effects of Succinylcholine. *Risk C: Monitor*

Suvorexant: CNS Depressants may increase CNS depressant effects of Suvorexant. *Risk C: Monitor*

Tasimelteon: CNS Depressants may increase CNS depressant effects of Tasimelteon. *Risk C: Monitor*

Tetrabenazine: CNS Depressants may increase CNS depressant effects of Tetrabenazine. *Risk C: Monitor*

Thalidomide: CNS Depressants may increase CNS depressant effects of Thalidomide. *Risk X: Avoid*

Thioridazine: QT-prolonging Agents (Moderate Risk) may increase QTc-prolonging effects of Thioridazine. *Risk X: Avoid*

Thiothixene: CNS Depressants may increase CNS depressant effects of Thiothixene. *Risk C: Monitor*

TiaGABine: CNS Depressants may increase CNS depressant effects of TiaGABine. *Risk C: Monitor*

TiZANidine: CNS Depressants may increase CNS depressant effects of TiZANidine. *Risk C: Monitor*

Tolcapone: Blood Pressure Lowering Agents may increase hypotensive effects of Tolcapone. *Risk C: Monitor*

Tolcapone: CNS Depressants may increase CNS depressant effects of Tolcapone. *Risk C: Monitor*

Topiramate: CNS Depressants may increase CNS depressant effects of Topiramate. CNS Depressants may increase adverse/toxic effects of Topiramate. Specifically, cognitive dysfunction and neuropsychiatric adverse effects may be increased. *Risk C: Monitor*

Tradipitant: CNS Depressants may increase CNS depressant effects of Tradipitant. *Risk C: Monitor*

TraZODone: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trifluoperazine: CNS Depressants may increase CNS depressant effects of Trifluoperazine. *Risk C: Monitor*

Trimeprazine: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Trimethobenzamide: CNS Depressants may increase CNS depressant effects of Trimethobenzamide. Management: Avoid coadministration of trimethobenzamide with other CNS depressants if possible. If coadministration cannot be avoided, monitor patients for CNS adverse effects (eg, depression, disorientation, seizures). *Risk D: Consider Therapy Modification*

Trimipramine: CNS Depressants may increase CNS depressant effects of Trimipramine. *Risk C: Monitor*

Triprolidine: CNS Depressants may increase CNS depressant effects of Triprolidine. *Risk C: Monitor*

Valerian: May increase CNS depressant effects of CNS Depressants. *Risk C: Monitor*

Valproic Acid and Derivatives: CNS Depressants may increase CNS depressant effects of Valproic Acid and Derivatives. *Risk C: Monitor*

Vigabatrin: CNS Depressants may increase CNS depressant effects of Vigabatrin. *Risk C: Monitor*

Zaleplon: CNS Depressants may increase CNS depressant effects of Zaleplon. Management: Coadministration of zaleplon with other sedative-hypnotics at bedtime or in the middle of the night is not recommended. If zaleplon is coadministered with other CNS depressants at other times, monitor for CNS depression; dose reductions may be required. *Risk D: Consider Therapy Modification*

Ziconotide: CNS Depressants may increase CNS depressant effects of Ziconotide. *Risk C: Monitor*

Zolpidem: CNS Depressants may increase CNS depressant effects of Zolpidem. Management: Reduce the Intermezzo brand sublingual zolpidem adult dose to 1.75 mg for men who are also receiving other CNS depressants. No such dose change is recommended for women. Avoid use with other CNS depressants at bedtime; avoid use with alcohol. *Risk D: Consider Therapy Modification*

Zonisamide: CNS Depressants may increase CNS depressant effects of Zonisamide. *Risk C: Monitor*

Zopiclone: CNS Depressants may increase CNS depressant effects of Zopiclone. *Risk C: Monitor*

Zuclopenthixol: May increase CNS depressant effects of CNS Depressants. Management: Avoid use of high-dose hypnotics or other CNS depressants together with zuclopenthixol. *Risk D: Consider Therapy Modification*

Zuranolone: May increase CNS depressant effects of CNS Depressants. Management: Consider alternatives to the use of zuranolone with other CNS depressants or alcohol. If combined, consider a zuranolone dose reduction and monitor patients closely for increased CNS depressant effects. *Risk D: Consider Therapy Modification*

Pregnancy Considerations

Based on animal data, repeated or prolonged use of general anesthetic and sedation medications that block N-methyl-D- aspartate (NMDA) receptors and/or potentiate gamma-aminobutyric acid (GABA) activity, may affect brain development. Evaluate benefits and potential risks of fetal exposure to desflurane when duration of surgery is expected to be >3 hours (Olutoye 2018).

Use of desflurane in obstetric anesthesia has been described (Boat 2010; Karaman 2006; Patel 1995). However, other agents are more commonly used (ACOG 209 2019; Devroe 2015). Maternal exposure should be minimized due to dose dependent uterine relaxation and fetal depression (Devroe 2015).

The ACOG recommends that pregnant women should not be denied medically necessary surgery, regardless of trimester. If the procedure is elective, it should be delayed until after delivery (ACOG 775 2019).

Breastfeeding Considerations

It is not known if desflurane is present in breast milk.

The manufacturer recommends that caution be exercised when administering desflurane to breast-feeding females; however, based on pharmacokinetic properties, use of desflurane for the maintenance of anesthesia in breastfeeding women is acceptable (Dalal 2014). The Academy of Breast Feeding Medicine recommends postponing elective surgery until milk supply and breastfeeding are established. Milk should be expressed ahead of surgery when possible. In general, when the child is healthy and full term, breastfeeding may resume, or milk may be expressed once the mother is awake and in recovery. For children who are at risk for apnea, hypotension, or hypotonia, milk may be saved for later use when the child is at lower risk (ABM [Reece-Stremtan 2017]).

Monitoring Parameters

Blood pressure, heart rate and rhythm, temperature, oxygen saturation, end-tidal CO_2 and end-tidal desflurane concentrations should be monitored prior to and throughout anesthesia; adverse respiratory symptoms in pediatric patients (≤6 years); signs and symptoms of airway narrowing in children with asthma or recent upper respiratory infection

Mechanism of Action

Although not completely defined, it is thought that desflurane enhances inhibitory postsynaptic channel activity and inhibits excitatory synaptic activity resulting in general anesthesia.

Pharmacokinetics (Adult Data Unless Noted)

Onset of action: 1 to 2 minutes

Duration: Emergence time: Depends on blood concentration when desflurane is discontinued

The rate of change of anesthetic concentration in the lung is more rapid with desflurane because of its low blood/gas solubility (0.42), which is similar to nitrous oxide.

Metabolism: Hepatic (0.02%) to trifluoroacetate (negligible) and inorganic fluoride

Excretion: Exhaled gases

Patient Counseling Points

(For additional information see ”Desflurane: Patient drug information”)

What is this drug used for?

• It is used to put you to sleep for surgery.

• It is used to cause sleep during a procedure.

All drugs may cause side effects. However, many people have no side effects or only have minor side effects. Call your doctor or get medical help if any of these side effects or any other side effects bother you or do not go away:

• Upset stomach or throwing up

• Headache

• Cough

• More saliva

• Throat irritation

WARNING/CAUTION: Even though it may be rare, some people may have very bad and sometimes deadly side effects when taking a drug. Tell your doctor or get medical help

right away if you have any of the following signs or symptoms that may be related to a very bad side effect:

• High potassium levels like a heartbeat that does not feel normal; feeling confused; feeling weak, lightheaded, or dizzy; feeling like passing out; numbness or tingling; or shortness of breath

• High or low blood pressure like very bad headache or dizziness, passing out, or change in eyesight

• Trouble breathing, slow breathing, or shallow breathing

• Slow heartbeat

• Muscle stiffness

• Blue or gray color of the skin, lips, nail beds, fingers, or toes

• Not able to pass urine or change in how much urine is passed

• Feeling agitated

• Malignant hyperthermia like a fast heartbeat, fast breathing, fever, or spasm or stiffness of the jaw muscles

• A type of abnormal heartbeat (prolonged QT interval) like a fast or abnormal heartbeat, or if you pass out

• Liver problems like dark urine, tiredness, decreased appetite, upset stomach or stomach pain, light-colored stools, throwing up, or yellow skin or eyes

• Signs of an allergic reaction, like rash; hives; itching; red, swollen, blistered, or peeling skin with or without fever; wheezing; tightness in the chest or throat; trouble breathing, swallowing, or talking; unusual hoarseness; or swelling of the mouth, face, lips, tongue, or throat.

Note: This is not a comprehensive list of all side effects. Talk to your doctor if you have questions.

Consumer Information Use and Disclaimer: This information should not be used to decide whether or not to take this medicine or any other medicine. Only the healthcare provider has the knowledge and training to decide which medicines are right for a specific patient. This information does not endorse any medicine as safe, effective, or approved for treating any patient or health condition. This is only a limited summary of general information about the medicine's uses from the patient education leaflet and is not intended to be comprehensive. This limited summary does NOT include all information available about the possible uses, directions, warnings, precautions, interactions, adverse effects, or risks that may apply to this medicine. This information is not intended to provide medical advice, diagnosis or treatment and does not replace information you receive from the healthcare provider. For a more detailed summary of information about the risks and benefits of using this medicine, please speak with your healthcare provider and review the entire patient education leaflet.

Brand Names: International

International Brand Names by Country

For country code abbreviations (show table)

- (AE) United Arab Emirates: Suprane;
- (AR) Argentina: Suprane;
- (AT) Austria: Desfluran piramal | Suprane;
- (AU) Australia: Desflurane Sandoz;
- (BE) Belgium: Suprane;
- (BR) Brazil: Desforane;
- (CL) Chile: Desflurano | Suprane;
- (CN) China: Suprane;
- (CO) Colombia: Suprane;

- (CR) Costa Rica: Suprane;
- (CZ) Czech Republic: Suprane;
- (DE) Germany: Desfluran piramal | Suprane;
- (EC) Ecuador: Suprane;
- (EE) Estonia: Suprane;
- (EG) Egypt: Suprane;
- (ES) Spain: Suprane;
- (FI) Finland: Suprane;
- (FR) France: Suprane;
- (GB) United Kingdom: Suprane;
- (GR) Greece: Suprane;
- (HK) Hong Kong: Suprane;
- (HU) Hungary: Suprane;
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- (IT) Italy: Desflurano Piramal | Suprane;
- (JO) Jordan: Suprane;
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- (KR) Korea, Republic of: Suprane;
- (KW) Kuwait: Suprane;
- (LB) Lebanon: Suprane;
- (LT) Lithuania: Suprane;
- (LV) Latvia: Suprane;
- (MX) Mexico: Suprane;
- (MY) Malaysia: Suprane;
- (NO) Norway: Suprane;
- (NZ) New Zealand: Suprane;
- (PA) Panama: Suprane;
- (PH) Philippines: Suprane;
- (PL) Poland: Desfluran piramal | Suprane;
- (PR) Puerto Rico: Suprane;
- (PT) Portugal: Suprane;
- (QA) Qatar: Suprane;
- (RU) Russian Federation: Longradax | Suprane;
- (SA) Saudi Arabia: Suprane;
- (SE) Sweden: Suprane;
- (SG) Singapore: Suprane;
- (SK) Slovakia: Suprane;
- (TH) Thailand: Suprane;
- (TN) Tunisia: Suprane;
- (TR) Turkey: Suprane;
- (TW) Taiwan: Suprane;
- (UY) Uruguay: Suprane;
- (VE) Venezuela, Bolivarian Republic of: Suprane;
- (ZA) South Africa: Suprane | Torrane

REFERENCES

1. American College of Obstetricians and Gynecologists (ACOG). ACOG Practice Bulletin No. 209: Obstetric analgesia and anesthesia. *Obstet Gynecol*. 2019;133(3):e208-e225. [PubMed 30801474]

2. American College of Obstetricians and Gynecologists (ACOG). ACOG committee opinion no. 775: nonobstetric surgery during pregnancy. *Obstet Gynecol*. 2019;133(4):e285-e286. [PubMed

30913200]

3. Boat A, Mahmoud M, Michelfelder EC, et al. Supplementing desflurane with intravenous anesthesia reduces fetal cardiac dysfunction during open fetal surgery. *Paediatr Anaesth*. 2010;20(8):748-756. doi:10.1111/j.1460-9592.2010.03350.x [PubMed 20670239]
4. Chalasani NP, Maddur H, Russo MW, Wong RJ, Reddy KR; Practice parameters committee of the American College of Gastroenterology. ACG clinical guideline: diagnosis and management of idiosyncratic drug-induced liver injury. *Am J Gastroenterol*. 2021;116(5):878-898. doi:10.14309/ajg.0000000000001259 [PubMed 33929376]
5. Chung PC, Chiou SC, Lien JM, Li AH, Wong CH. Reproducible hepatic dysfunction following separate anesthesia with sevoflurane and desflurane. *Chang Gung Med J*. 2003;26(5):357-362. [PubMed 12934853]
6. Côté G, Bouchard S. Hepatotoxicity after desflurane anesthesia in a 15-month-old child with Mobius syndrome after previous exposure to isoflurane. *Anesthesiology*. 2007;107(5):843-845. doi:10.1097/01.anes.0000287196.15665.16 [PubMed 18073559]
7. Dalal PG, Bosak J, Berlin C. Safety of the breast-feeding infant after maternal anesthesia. *Paediatr Anaesth*. 2014;24(4):359-371. doi:10.1111/pan.12331 [PubMed 24372776]
8. Devroe S, Van de Velde M, Rex S. General anesthesia for caesarean section. *Curr Opin Anaesthesiol*. 2015;28(3):240-246. doi:10.1097/ACO.0000000000000185 [PubMed 25827280]
9. Food and Drug Administration. FDA Drug Safety Communication: FDA review results in new warnings about using general anesthetics and sedation drugs in young children and pregnant women. https://www.fda.gov/drugs/drug-safety-and-availability/fda-drug-safety-communication-fda-approves-label-changes-use-general-anesthetic-and-sedation-drugs. Updated April 27, 2017.
10. Han DW, Park K, Jang SB, Kern SE. Modeling the effect of sevoflurane on corrected QT prolongation: a pharmacodynamic analysis. *Anesthesiology*. 2010;113(4):806-811. [PubMed 20808206]
11. Kang J, Reynolds WP, Chen XL, et al. Mechanisms underlying the QT interval-prolonging effects of sevoflurane and its interactions with other QT-prolonging drugs. *Anesthesiology*. 2006;104(5):1015-1022. [PubMed 16645454]
12. Karaman S, Akercan F, Aldemir O, Terek MC, Yalaz M, Firat V. The maternal and neonatal effects of the volatile anaesthetic agents desflurane and sevoflurane in caesarean section: a prospective, randomized clinical study. *J Int Med Res*. 2006;34(2):183-192. [PubMed 16749414]
13. Nakao S, Hatano K, Sumi C, et al. Sevoflurane causes greater QTc interval prolongation in elderly patients than in younger patients. *Anesth Analg*. 2010;110(3):775-779. [PubMed 20185656]
14. Olutoye OA, Baker BW, Belfort MA, Olutoye OO. Food and Drug Administration warning on anesthesia and brain development: implications for obstetric and fetal surgery. *Am J Obstet Gynecol*. 2018;218(1):98-102. [PubMed 28888583]
15. Page RL 2nd, O'Bryant CL, Cheng D, et al; American Heart Association Clinical Pharmacology and Heart Failure and Transplantation Committees of the Council on Clinical Cardiology; Council on Cardiovascular Surgery and Anesthesia; Council on Cardiovascular and Stroke Nursing; and Council on Quality of Care and Outcomes Research. Drugs That May Cause or Exacerbate Heart Failure: A Scientific Statement From the American Heart Association. *Circulation*. 2016;134(6):e32-e69. [PubMed 27400984]
16. Patel SS, Goa KL. Desflurane. A review of its pharmacodynamic and pharmacokinetic properties and its efficacy in general anaesthesia. *Drugs*. 1995;50(4):742-767. [PubMed 8536556]
17. Reece-Stremtan S, Campos M, Kokajko L; Academy of Breastfeeding Medicine. ABM clinical protocol #15: analgesia and anesthesia for the breastfeeding mother, revised 2017. *Breastfeed Med*. 2017;12(9):500-506. [PubMed 29624435]

18. Suprane (desflurane) [prescribing information]. Deerfield, IL: Baxter Healthcare Corporation; October 2025.

19. Suprane (desflurane) [product monograph]. Mississauga, Ontario, Canada: Baxter Corporation; July 2023.

20. Weiskopf RB, Eger EI 2nd, Noorani M, et al. Fentanyl, Esmolol, and Clonidine Blunt the Transient Cardiovascular Stimulation Induced by Desflurane in Humans. *Anesthesiology*. 1994;81(6):1350-1355. [PubMed 7992902]

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